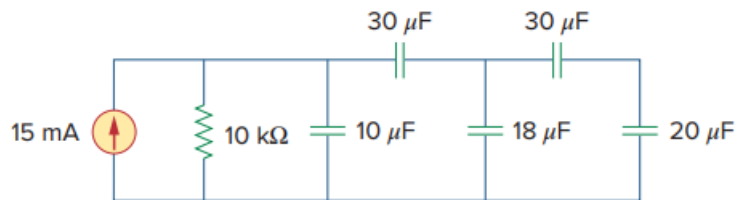


Problem

In the circuit shown in Fig. 6.58 assume that the capacitors were initially uncharged and that the current source has been connected to the circuit long enough for all the capacitors to reach steady state (no current flowing through the capacitors). Determine the voltage across each capacitor and the energy stored in each.

Fig. 6.58:



Step-by-step solution

Step 1 of 13

Refer to the circuit in Figure 6.58 in the textbook.

Under the steady state condition all the capacitor acts as open circuit and the circuit is reduced to as shown in Figure 1.

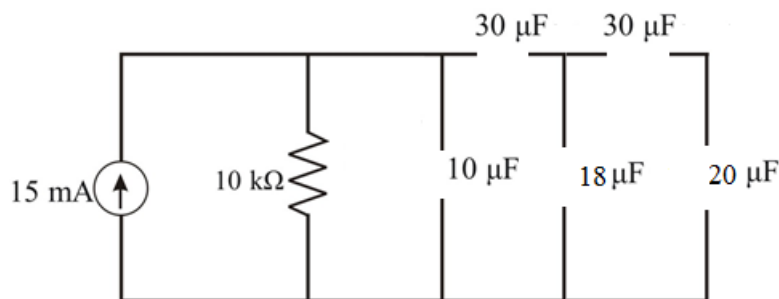


Figure 1

Step 2 of 13

From the circuit in Figure 1, under steady state condition the voltage across the $10\text{ k}\Omega$ resistor is,

$$\begin{aligned} v_{10\text{ k}\Omega} &= (15\text{ mA})(10\text{ k}\Omega) \\ &= 150\text{ V} \end{aligned}$$

From the circuit in Figure 1, the voltage across $10\text{ }\mu\text{F}$ capacitor is same as the voltage across the $10\text{ k}\Omega$ resistor.

Hence, the voltage across $10\text{ }\mu\text{F}$ capacitor is,

$$v_{10\text{ }\mu\text{F}} = 150\text{ V}$$

Thus, the voltage across $10\text{ }\mu\text{F}$ capacitor is $\boxed{150\text{ V}}$.

Step 3 of 13

The energy stored in $10\ \mu\text{F}$ capacitor is,

$$W_{10\ \mu\text{F}} = \frac{1}{2} C v_{10\ \mu\text{F}}^2$$

Substitute 150 V for $v_{10\ \mu\text{F}}$ and $10\ \mu\text{F}$ for C.

$$\begin{aligned} W_{10\ \mu\text{F}} &= \frac{1}{2} (10 \times 10^{-6}) (150)^2 \\ &= 0.1125\ \text{J} \end{aligned}$$

Thus, the energy stored in $10\ \mu\text{F}$ capacitor is 0.1125 J.

Step 4 of 13

From the circuit, the $20\ \mu\text{F}$ and $30\ \mu\text{F}$ capacitors are in series, and the combination is in series with $18\ \mu\text{F}$. So equivalent capacitance is,

$$\begin{aligned} &= 18\ \mu + \frac{20\ \mu \times 30\ \mu}{20\ \mu + 30\ \mu} \\ &= 18\ \mu + 12\ \mu \\ &= 30\ \mu\text{F} \end{aligned}$$

The modified circuit is shown in Figure 2.

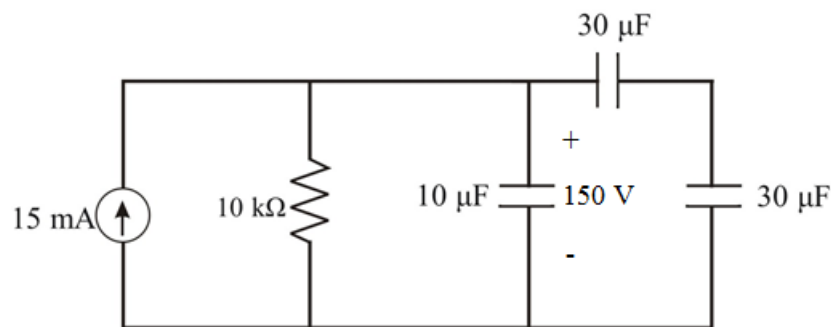


Figure 2

[Comments \(1\)](#)



Anonymous

Should read "...the 20 and 30 capacitors are in series, and the combination is in PARALLEL with 18..."

Step 5 of 13

By voltage division rule the voltage across the $30\ \mu\text{F}$ capacitor is,

$$\begin{aligned} v_{30\ \mu\text{F}} &= 150 \times \frac{30\ \mu\text{F}}{30\ \mu\text{F} + 30\ \mu\text{F}} \\ &= 150 \times \frac{1}{2} \\ &= 75\ \text{V} \end{aligned}$$

Thus, the voltage across $30\ \mu\text{F}$ capacitor is 75 V.

Step 6 of 13

The energy stored in $30\ \mu\text{F}$ capacitor is,

$$W_{30\ \mu\text{F}} = \frac{1}{2} C v_{30\ \mu\text{F}}^2$$

Substitute 75 V for $v_{30\ \mu\text{F}}$ and $30\ \mu\text{F}$ for C.

$$\begin{aligned} W_{30\ \mu\text{F}} &= \frac{1}{2} (30 \times 10^{-6}) (75)^2 \\ &= 84.375\ \text{mJ} \end{aligned}$$

Thus, the energy stored in $30\ \mu\text{F}$ capacitor is $84.375\ \text{mJ}$.

Step 7 of 13

The voltage across the $18\ \mu\text{F}$ capacitor is,

$$\begin{aligned} v_{18\ \mu\text{F}} &= v_{30\ \mu\text{F}} \\ &= 150 \times \frac{30\ \mu\text{F}}{30\ \mu\text{F} + 30\ \mu\text{F}} \\ &= 150 \times \frac{1}{2} \\ &= 75\ \text{V} \end{aligned}$$

Thus, the voltage across $18\ \mu\text{F}$ capacitor is $75\ \text{V}$.

Step 8 of 13

The energy stored in $18\ \mu\text{F}$ capacitor is,

$$W_{18\ \mu\text{F}} = \frac{1}{2} C v_{18\ \mu\text{F}}^2$$

Substitute 75 V for $v_{18\ \mu\text{F}}$ and $18\ \mu\text{F}$ for C.

$$\begin{aligned} W_{18\ \mu\text{F}} &= \frac{1}{2} (18 \times 10^{-6}) (75)^2 \\ &= 50.625\ \text{mJ} \end{aligned}$$

Thus, the energy stored in $18\ \mu\text{F}$ capacitor is $50.625\ \text{mJ}$.

Step 9 of 13

Consider the circuit in Figure 3.

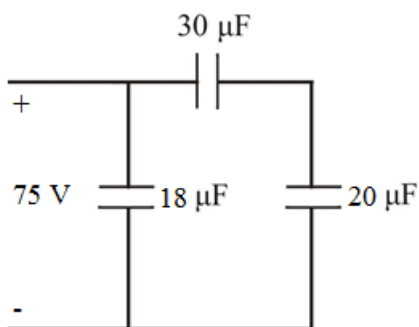


Figure 3

Step 10 of 13

By voltage division rule the voltage across the $30\ \mu\text{F}$ capacitor is,

$$\begin{aligned}v_{30\ \mu\text{F}} &= 75 \times \frac{20\ \mu\text{F}}{30\ \mu\text{F} + 20\ \mu\text{F}} \\&= 75 \times \frac{2}{5} \\&= 30\ \text{V}\end{aligned}$$

Thus, the voltage across $30\ \mu\text{F}$ capacitor is $\boxed{30\ \text{V}}$.

Step 11 of 13

The energy stored in $30\ \mu\text{F}$ capacitor is,

$$W_{30\ \mu\text{F}} = \frac{1}{2} C v_{30\ \mu\text{F}}^2$$

Substitute 30 V for $v_{30\ \mu\text{F}}$ and $30\ \mu\text{F}$ for C.

$$\begin{aligned}W_{30\ \mu\text{F}} &= \frac{1}{2} (30 \times 10^{-6}) (30)^2 \\&= 13.5\ \text{mJ}\end{aligned}$$

Thus, the energy stored in $30\ \mu\text{F}$ capacitor is $\boxed{13.5\ \text{mJ}}$.

Step 12 of 13

By voltage division rule the voltage across the $20\ \mu\text{F}$ capacitor is,

$$\begin{aligned}v_{20\ \mu\text{F}} &= 75 \times \frac{30\ \mu\text{F}}{30\ \mu\text{F} + 20\ \mu\text{F}} \\&= 75 \times \frac{3}{5} \\&= 45\ \text{V}\end{aligned}$$

Thus, the voltage across $20\ \mu\text{F}$ capacitor is $\boxed{45\ \text{V}}$.

Step 13 of 13

The energy stored in $20\ \mu\text{F}$ capacitor is,

$$W_{20\ \mu\text{F}} = \frac{1}{2} C v_{20\ \mu\text{F}}^2$$

Substitute 45 V for $v_{20\ \mu\text{F}}$ and $20\ \mu\text{F}$ for C.

$$\begin{aligned}W_{20\ \mu\text{F}} &= \frac{1}{2} (20 \times 10^{-6}) (45)^2 \\&= 20.25\ \text{mJ}\end{aligned}$$

Thus, the energy stored in $20\ \mu\text{F}$ capacitor is $\boxed{20.25\ \text{mJ}}$.